Gwangho Kim

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Research Interests

My research interests lie in the theoretical understanding of deep learning and generative models. Currently, I focus on memorization and generalization in diffusion models—specifically, how they learn from finite training data and generalize beyond the samples they have seen. Building on this understanding, I aim to develop reliable, efficient, and high-performance generative models across a range of domains.

Education

Hanyang University, Seoul, South Korea *Mar. 2025 - Present*

M.S. in Computer Science ***GPA 97.9/100***

- Research interests: Machine Learning and Generative Models
- GPA: 4.29 / 4.5

Hanyang University, Seoul, South Korea *Mar. 2018 - Sep. 2024*

B.S. in Computer Science & Mathematics ***GPA 94.7/100***

- Double Major in Computer Science and Mathematics
- GPA: 3.97 / 4.5
- Military Service (2019 - 2020)

Publications & Preprints

Memorization and Malign Generalization in Conditional Diffusion Models with Random Features
Under review. arXiv:2610.11288.
Gwangho Kim, Sungyoon Lee. 2026.

Parallel Tempering Initial Sampling in Inference-Time Reward Alignment
ICML 2026 Workshop on Structured Probabilistic Inference & Generative Modeling (SPIGM).
Myeongjun Oh, **Gwangho Kim**, Sungyoon Lee. 2026.

Localizing Memorized Regions in Diffusion Models via Coordinate-Wise Curvature Differences
International Conference on Machine Learning (**ICML 2026**).
Gwangho Kim, Sungyoon Lee. 2026.

Honors & Awards

Fall 2023 **Hanyang Brain Academic Excellence Scholarship**, Hanyang University
2018, 2021 **Freshmen Merit-Based Scholarship**, Hanyang University

Teaching Experience

Fall 2025 **INTRODUCTION TO MACHINE LEARNING THEORY**, Teaching Assistant
Spring 2025 **AI Thesis Research**, Teaching Assistant

Academic Service

Reviewer: ICML SPIGM Workshop (2026)